legio

_travelers guide
_for navigating the legio starsystem

23	5	28
noise engineering firmwares	from the community	firmwares on one sheet

[enc] press [enc] hold › then noon = 12 o'clock cw / ccw = clockwise / counter

one 6 hp body · many minds · every control below is named by its tag on the panel drawing

labels from the noise engineering printable overlays · firmware data from each developer, 2026.10

orange = hold

01 the platform

lg_01 · fig.03

enc	encoder + cv jack (−2 to +5 v) · some firmwares add a tap
1 2	two knobs, each with a cv jack, 0 to +5 v
s1 s2	two 3-way switches, read up · mid · down
trig	trigger jack, about 1.6 v: hit, sync, fsu, mute…
in out	stereo · L feeds R when R is empty
LEDs	boot colours identify the firmware
size	6 hp · 1.5 in deep · 140 mA +12 v · 22 mA −12 v

02 swapping firmware

lg_02

hot swap

official ne firmware · no button combo

1. plug usb into the module (port on the back) and your computer · **it can stay in the case, powered**
2. open the noise engineering firmware swap page, choose legio, click **connect**, then **swap** · about half the old time

route a usb cable out of the case to swap mid-patch · usb data cable + chrome, edge or opera · not firefox or safari

first time

one last swap the old way

old way	power off, unscrew the legio, unplug its power, usb in, click **help** on the firmware page · any ne firmware turns hot swap on · swap the overlay to match

community firmware

built from source

flash	daisy patch sm in dfu mode (hold BOOT, tap RESET) · make program-dfu or dfu-util
go back	no hot swap from community firmware · back to ne the old way, with the button combo

06 community

lg_06 · continued

jawari legio

blue nautilus · github

tanpura drone: four strings strummed in turn, after the vcv module

enc tuning · push = coarse | 1 jawari shape | 2 lpf / string 1
s1 knob 2 lpf · string 1 | s2 formants on · off | trig strum

mono, same on both outs · pitch + semitone are saved

rotary legio

gugenet · github

rotary speaker: horn + drum rotors, doppler, three cabinets

enc tap = max · balance · drive | 1 doppler | 2 mic distance
s1 speed max · cv · brake | s2 cab open · classic · dark | trig max speed | in in

hold the encoder 0.8 s = bypass · enc cv jack sweeps speed

03 oscillators + voices

lg_03 · 9 firmwares

virt iter legio

enter virt iter sector · manual

multi-algorithm stereo oscillator · a launch firmware

enc pitch | 1 flavor | 2 tang
s1 harm · sawx · bass | s2 II · I · 0 | trig sync · in pm

sinc legio

enter sinc sector · manual

stereo phase-modulation oscillator

enc pitch | 1 morph | 2 timbre
s1 warp · op · fold | s2 II · I · 0 | trig sync · in pm

gator legio

enter gator sector · manual

supersaw phase-mod oscillator, stereo pm ins

enc pitch | 1 detune | 2 saw mod
s1 saw count | s2 stereoness | trig 8va · in pm

at legio

enter at sector · manual

lfsr lo-fi oscillator after the ataraxic translatron

enc pitch | 1 tone | 2 mod
s1 mode env · rm · mix | s2 II · I · 0 | trig sync · in in

plus legio

enter plus sector · manual

full additive voice with envelope + lfo

enc pitch | 1 time | 2 timbre
s1 mod + · off · − | s2 lfo env · on · off | trig hit · in pm

03 oscillators + voices

lg_03 · continued

unda legio

enter unda sector · manual

full phase-mod voice, built for chaining

enc pitch | 1 time | 2 timbre
s1 mod +1 · 0 · −1 | s2 mode adsr · asr · ar | trig hit · in mix

reso legio

enter reso sector · manual

karplus-strong string voice + resonator

enc pitch | 1 decay | 2 brightness
s1 body banjo · violin · guitar | s2 chorus II · I · 0 | trig hit · in excite

marimba legio

enter marimba sector · manual

physical-model resonator voice + processor

enc pitch | 1 decay | 2 tone
s1 type tube · metal · wood | s2 mode ext · bow · hit | trig hit · in excite

morbus legio

enter morbus sector · manual

stereo noise generator

enc frequency | 1 harsh · in: fwr | 2 color · in: clip
s1 wtf · rough · none | s2 II · I · 0 | trig mute · in in

04 drums

lg_04 · 7 firmwares

kick legio

enter kick sector · manual

analog-style kick that ducks the audio at its ins

enc pitch / mute | 1 decay | 2 impact
s1 tone | s2 sat | trig hit · in duck

snare legio

enter snare sector · manual

physical-model snare

enc pitch / mute | 1 decay | 2 noise
s1 impact | s2 wire | trig hit · in mix

tom legio

enter tom sector · manual

toms, lasers, zaps, mid-range hits

enc pitch / mute | 1 decay | 2 punch
s1 tone | s2 sat | trig hit · in mix

cymbal legio

enter cymbal sector · manual

granular metal for hats, rides + crashes

enc morph / mute | 1 decay | 2 tone
s1 type ride · crash · hat | s2 alloy | trig hit · in mix

bos legio

enter bos sector · manual

more cowbell, stereo pm ins

enc pitch / mute | 1 decay | 2 tone
s1 mode fm · add · x0x | s2 tilt | trig hit · in fm

klang legio

enter klang sector · manual

zappy percussion after basimilus

enc pitch / mute | 1 decay | 2 impact
s1 harm | s2 spread | trig hit · in mix

tymp legio

enter tymp sector · manual

compact stereo digital drum

enc pitch | 1 decay | 2 wackitude
s1 mode boing · cat · boot | s2 tang | trig hit · in quack

06 community

lg_06 · 5 open firmwares

sawstack legio

carl tashian · github · altstrument

five-voice supersaw, no envelope · also an au / vst3 plugin · mit

enc fine pitch · push = coarse | 1 detune | 2 morph
s1 mode stack · rich · sub | s2 width mono · stereo · wide | trig hard sync

stutterer legio

carl tashian · github · altstrument

stutter › tape delay › dj filter › soft clip, clock-synced · mit

enc tape speed · tap = stutter | 1 subdivision | 2 dj filter
s1 clock ×2 · =1 · ÷2 | s2 wet 100 · 50 · 0 % | trig clock · in in

drifter legio

carl tashian · github · altstrument

random bézier auto-panner, crossfader or cv panner · mit

enc rate · tap = curve | 1 center | 2 depth
s1 mode pan · xfade · cv pan | s2 edge clip · fold · wrap | trig clock · in in

05 effects

lg_05 · 7 firmwares

librae legio

enter librae sector · manual

dynamics: compress, limit, expand, gate · a launch firmware

enc color | 1 ceiling | 2 room
s1 tilt high · off · low | s2 cut max · some · none | trig fsu · in in

chasma legio

enter chasma sector · manual

mid / side compression + limiting

enc distortion | 1 side limit | 2 mid limit
s1 mid comp | s2 side comp | trig fsu · in in

ampla legio

enter ampla sector · manual

multimode filter + gate, tube-style drive

enc freq | 1 a/d | 2 time
s1 howl · bite · calm | s2 hp · bp · lp | trig hit · in in

roucha legio

enter roucha sector · manual

stereo multimode filter with folder

enc frequency · tap = bypass | 1 resonance | 2 fold
s1 all · post · pre | s2 hp · bp · lp | trig bypass · in in

bitcrush legio

enter bitcrush sector · manual

pitch-tracking downsampler, bits + dither

enc rate / bypass | 1 bits | 2 dither
s1 env in · out · off | s2 lpf off · high · low | trig bypass · in in

megafold legio

enter megafold sector · manual

stereo wavefolder + shaper with chorus

enc envelope | 1 gain | 2 shape
s1 chorus lots · some · none | s2 speed fast · mid · slow | trig fsu · in in

ovis legio

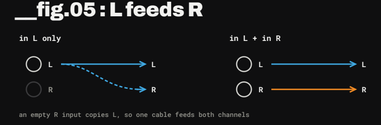
enter ovis sector · manual

granular shepard-tone processor

enc pitch | 1 count | 2 length
s1 fade | s2 layers | trig hit · in in

__fig.05 : L feeds R

in · out



an empty R input copies L, so one cable feeds both channels

__fig.03 : tagged panel

the tags every card uses

[enc] __encoder : turn · tap in some firmwares
[1] __top knob : the first big knob
[2] __bottom knob : the second big knob
[s1] __left switch : up · mid · down
[s2] __right switch : up · mid · down
[trig] __jack : hit / sync / fsu, about 1.6 v
[cv] __enc · 1 · 2 : a cv jack for each
[in] __in L / R : L feeds R if alone
[out] __out L / R : stereo out
[LED] __two lights : boot colours = firmware



reading a card · kick legio as an example

kick legio enter kick sector · manual

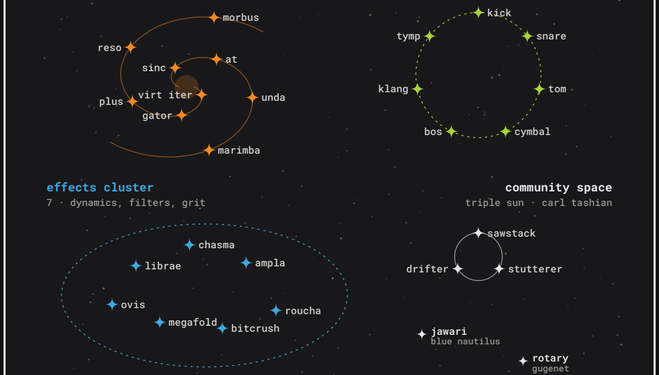
analog-style kick that ducks the audio at its ins

enc pitch / mute | 1 decay | 2 impact
s1 tone | s2 sat | trig hit · in duck

the encoder and both knobs also have a cv jack under the same name · switch positions read top to bottom · "pitch / mute" = turn for pitch, tap to mute

__fig.04 : the star map

28 firmwares by family



rogue stars: one firmware per maker